Mobile Receipt Extraction API

A focused integration guide for mobile developers: endpoints, multipart upload, response data, and error handling.

Integration boundary
The mobile application calls the NestJS API. The internal FastAPI AI service is not exposed directly to mobile clients.

Mode	Endpoint	Authentication	Persistence
Signed in	POST /receipts/extract	Firebase ID token	Receipt and items saved
Guest	POST /guest/receipts/extract	None	Not saved

1. Request contract

Rule	Value
HTTP format	multipart/form-data
File field	image
Formats	JPG, JPEG, PNG, WEBP, PDF
Maximum size	10 MB
Guest limit	5 requests per 60 seconds

Important multipart rule
Do not manually set the Content-Type header. Let the HTTP client generate it with the correct multipart boundary.

Authenticated request

```
POST /receipts/extract
Authorization: Bearer <FIREBASE_ID_TOKEN>
Content-Type: multipart/form-data

image: <binary file>
```

Guest request

```
POST /guest/receipts/extract
Content-Type: multipart/form-data

image: <binary file>
```

2. Response envelope

Every success response uses the same top-level shape:

```
type ApiResponse<T> = {
  status_code: number;
  title: string;
  body: string;
  data: T;
};
```

Receipt fields

Field	Type	Meaning
id	string	Saved UUID or temporary guest ID
documentType	string	Detected document category
language	string | null	Detected receipt language
country	string | null	Detected country
merchant	string | null	Merchant or business name
address	string | null	Merchant address
phone	string | null	Merchant phone number
invoiceNumber	string | null	Receipt or invoice reference
date	string | null	Detected transaction date
time	string | null	Detected transaction time
currency	string | null	Detected ISO currency code
subtotal	number | null	Original subtotal
tax	number | null	Original tax amount
total	number | null	Original total
convertedSubtotal	number | null	Subtotal in user's home currency
convertedTax	number | null	Tax in user's home currency
convertedTotal	number | null	Total in user's home currency
paymentMethod	string | null	Detected payment method
items	ReceiptItem[]	Extracted line items
otherText	string[] | null	Additional OCR text

Item fields

Field	Type	Meaning
id	string	Saved or temporary item identifier
name	string	Detected item name
description	string | null	Optional item details
quantity	number	Detected quantity
unitPrice	number | null	Original unit price
totalPrice	number | null	Original line total
convertedUnitPrice	number | null	Unit price in home currency
convertedTotalPrice	number | null	Line total in home currency

3. Example success response


```
{
  "status_code": 201,
  "title": "Created successfully",
  "body": "Created successfully",
  "data": {
    "id": "94770557-6a8a-46c6-a042-b560af2ba33d",
    "documentType": "receipt",
    "language": "fr",
    "country": "Tunisia",
    "merchant": "Carrefour",
    "address": "Tunis, Tunisia",
    "phone": null,
    "invoiceNumber": "INV-1024",
    "date": "2026-10-02",
    "time": "14:32",
    "currency": "TND",
    "subtotal": 18.5,
    "tax": 1.5,
    "total": 20,
    "convertedSubtotal": 18.5,
    "convertedTax": 1.5,
    "convertedTotal": 20,
    "paymentMethod": "card",
    "items": [{
      "id": "eb88a7cb-12da-4bd2-b838-075f36b7790e",
      "name": "Water",
      "description": null,
      "quantity": 2,
      "unitPrice": 1.5,
      "totalPrice": 3,
      "convertedUnitPrice": 1.5,
      "convertedTotalPrice": 3
    }],
    "otherText": []
  }
}
```


Nullability and OCR review
OCR fields can be missing or imperfect. The mobile UI must accept null values and should let the user review the extracted data. Converted values may be null or absent when conversion is unavailable.

4. React Native example

```
const formData = new FormData();

formData.append("image", {
  uri: imageUri,
  name: "receipt.jpg",
  type: "image/jpeg",
} as any);

const response = await fetch(
  `${API_BASE_URL}/receipts/extract`,
  {
    method: "POST",
    headers: {
      Authorization: `Bearer ${firebaseIdToken}`,
    },
    body: formData,
  }
);

const result = await response.json();

if (!response.ok) {
  throw new Error(
    result.title || result.body || "Receipt extraction failed"
  );
}

const receipt = result.data;
```

For guest mode, use /guest/receipts/extract and omit the Authorization header.

5. Error handling

```
{
  "status_code": 422,
  "title": "Receipt extraction failed",
  "body": "The uploaded image could not be processed",
  "data": null
}
```

Status	Mobile handling
400 / 422	Missing, invalid, or unreadable upload
401	Missing or expired Firebase token; request sign-in again
413	File exceeds the 10 MB limit
415	Unsupported file type
429	Guest rate limit or AI quota reached; retry later
502 / 503	OCR/AI service temporarily unavailable
500	Unexpected server error